

# Binary · IG · Practice

U02a · 24 written + 6 pen-first = 30 marks · Name: \_\_\_\_\_

Unless a real-paper citation is shown, questions are ours. Show working where requested; keep every digit of a stated register width.

## 1. Ours: watch the carry

(a) A four-wheel binary counter shows **0101**. Press +1 four times. Write the four resulting patterns in order. [2]

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(b) At the transition from 0111 to 1000, state the value of the wheel that turns on. [1]

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## 2. Ours: score registers

(a) Convert **53** to eight-bit binary. Show your working. [2]

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(b) Convert **170** to eight-bit binary. Show your working. [2]

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## 3. Ours: two states

A switch can be on or off. State why binary is suitable for representing these states. [1]

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## 4. Ours: read the stored values

(a) Give the denary value of **01010101**. Show the place values you add. [2]

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(b) Give the denary value of **11101011**. Show the place values you add. [2]

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## 5. Ours: the pool counter

A swimming-pool display shows 0300. Write the value in its **16-bit unsigned register**. [2]

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## 6. Ours: pattern count is not the last label

(a) State how many different patterns an eight-bit register can hold. [1]

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(b) State its largest unsigned denary value. [1]

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## 7. Ours: exam-shape rehearsal · visitor counter

A library records daily visitor counts in an 8-bit unsigned register. This is a fresh authored instance of the conversion/capacity question shape.

**(a)** There are 73 visitors on Monday and 198 on Tuesday. Show each count as an 8-bit binary register value. **[2]**

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**(b)** Give the largest denary value that can be stored in the 8-bit binary register. **[1]**

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**(c)** The library expands and needs to record up to 3000 visitors per day. Give the smallest number of bits that can store the denary value 3000. **[1]**

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## 8. Ours: addition and overflow

**(a)** Add the positive binary integers **11001000** and **01100100**. Show carries and the full mathematical sum, including any extra carry bit. **[2]**

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**(b)** Explain why the result causes unsigned overflow in an eight-bit register. Refer to the sum and the register's largest value. **[2]**

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## 9. Ours: six answers make a number · pen first [6]

 **Paper first, then the classroom cards.** No Python file is needed. Six cards are labelled by weights 1, 2, 4, 8, 16, 32. A number appears on a card exactly when that weight is used in its binary representation. Answer before trying the cards. These six marks are human marked.

(a) For the number **27**, write YES or NO in card order **1, 2, 4, 8, 16, 32**. Show the weights you used. [2]

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(b) A player answers **NO, YES, NO, YES, NO, YES** in that same order. Write the six-bit pattern (largest weight first) and its denary value. [2]

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(c) Add a seventh card. State its weight and the largest number the seven cards can identify when at least one answer is YES. [2]

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**Keyboard companion:** continue Karel 05 LightTrail and 06 TheLongHall, your first loops. This companion is outside the 30 marks.

**Above and beyond · outside the total.** A counter has three faces per wheel: 0, 1, 2. Predict what follows 022. Explain the denary values of 022 and 100 using place values. Why does a carry happen at a different moment?

**Additional space - label the question number.**

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